

HOLY CROSS, AK, USA

WMO: 703051

Lat: 62.183N

Lon: 159.767W

Elev: 6

StdP: 101.25

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 26521

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.7	-32.8	-36.2	0.1	-32.9	-33.7	0.2	-30.0	9.7	-14.0	8.8	-11.8	0.4	250	0.707

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	24.3	15.1	22.2	14.6	20.0	13.6	16.3	22.1	15.3	20.5	14.3	18.7	2.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	10.0	17.5	13.1	9.4	16.7	12.3	8.9	15.9	45.6	21.9	42.7	20.4	40.0	18.8	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.3	6.3	DB	-38.3	28.3	4.7	1.9	-41.7	29.7	-44.4	30.8	-47.1	31.9	-50.5	33.2	(4)
(5)			WB	-35.1	18.1	2.7	1.4	-37.0	19.2	-38.6	20.0	-40.1	20.8	-42.0	21.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.8	-15.0	-12.4	-11.5	-1.0	7.7	12.5	13.8	12.5	7.9	-0.1	-10.8	-13.6	(6)	
	DBStd	12.72	11.10	8.89	6.98	6.38	4.77	3.36	3.11	2.78	3.57	4.87	7.92	8.37	(7)	
	HDD10.0	4269	773	628	666	329	98	11	4	11	79	314	624	732	(8)	
	HDD18.3	6983	1032	861	924	579	329	181	144	182	314	572	874	990	(9)	
	CDD10.0	338	0	0	0	0	28	85	122	87	16	0	0	0	(10)	
	CDD18.3	9	0	0	0	0	1	4	4	0	0	0	0	0	(11)	
	CDH23.3	111	0	0	0	0	9	51	49	2	0	0	0	0	(12)	
	CDH26.7	18	0	0	0	0	1	10	8	0	0	0	0	0	(13)	
(14) Wind	WSAvg	2.5	2.8	3.3	3.1	2.6	2.4	2.2	2.1	1.9	2.1	2.3	2.7	2.5	(14)	
(15) Precipitation	PrecAvg	604	40	38	33	28	36	47	67	85	75	56	52	46	(15)	
	PrecMax	782	103	88	102	54	77	78	117	126	135	120	123	95	(16)	
	PrecMin	460	4	5	3	6	10	23	29	37	15	7	4	14	(17)	
	PrecStd	100	22	20	22	13	14	12	19	25	30	25	29	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	5.2	4.1	14.4	24.6	28.1	28.0	23.0	19.3	11.6	6.3	6.4	(19)	
		MCWB	3.1	2.7	1.3	6.9	13.7	16.3	17.1	15.4	12.7	8.1	3.6	3.2	(20)	
	2%	DB	4.2	3.2	2.5	12.0	21.8	24.6	24.3	20.5	16.7	9.1	3.4	3.7	(21)	
		MCWB	2.0	1.1	0.2	5.8	12.5	14.8	16.0	14.6	11.7	7.2	2.3	1.5	(22)	
	5%	DB	2.7	1.6	0.7	9.8	18.4	21.7	22.2	19.1	14.6	7.8	1.7	1.3	(23)	
		MCWB	0.9	0.0	-1.0	4.7	10.5	13.6	15.4	13.8	10.7	6.1	0.9	0.3	(24)	
	10%	DB	0.4	-0.3	-1.2	7.7	15.6	19.1	19.8	17.6	13.0	6.5	-0.1	-1.4	(25)	
		MCWB	-0.8	-1.3	-2.8	3.6	9.0	12.4	14.6	13.3	10.2	5.0	-0.7	-2.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	2.8	1.8	7.4	14.1	17.1	18.6	16.9	13.5	9.2	4.1	3.3	(27)	
		MCDB	5.9	4.8	3.3	13.3	23.6	25.8	24.8	20.4	17.3	10.8	4.8	4.9	(28)	
	2%	WB	2.2	1.4	0.6	6.0	12.7	15.6	16.9	15.5	12.2	7.6	2.3	1.7	(29)	
		MCDB	4.0	3.1	2.1	11.5	20.8	22.9	23.1	19.1	15.5	9.0	3.5	3.5	(30)	
	5%	WB	1.1	0.3	-0.9	4.9	10.9	14.1	15.7	14.6	11.2	6.4	0.9	0.3	(31)	
		MCDB	2.8	1.5	0.8	9.5	17.3	20.7	21.1	17.8	14.0	7.8	1.6	1.5	(32)	
	10%	WB	-0.7	-1.4	-2.8	3.8	9.5	12.8	14.8	13.8	10.4	5.1	-0.7	-2.0	(33)	
		MCDB	0.8	-0.2	-1.2	7.5	15.1	18.2	19.5	16.7	12.8	6.5	0.0	-1.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.1	9.1	9.2	9.9	9.0	7.9	7.7	7.1	4.9	5.6	6.0	(35)	
		MCDBR	7.3	6.4	8.4	11.3	14.1	13.6	12.9	10.6	10.0	5.7	4.8	6.6	(36)	
	5% WB	MCWBR	5.8	5.4	6.9	6.5	6.8	6.1	6.1	5.5	6.0	4.1	4.4	5.4	(37)	
		MCWBR	7.5	6.4	8.1	10.9	13.2	12.7	11.8	9.2	8.7	5.3	4.6	6.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.215	0.263	0.272	0.346	0.365	0.365	0.357	0.371	0.320	0.288	0.221	0.192	(40)	
		taud	2.036	2.229	2.268	2.131	2.330	2.384	2.451	2.416	2.535	2.431	2.053	1.906	(41)	
	Ebn at Noon		472	709	848	831	852	858	857	809	798	683	446	280	(42)	
		Edh at Noon	35	62	85	122	110	106	97	93	68	51	33	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.30	1.00	2.86	4.59	5.21	5.32	4.48	3.42	2.32	1.07	0.37	0.14	(44)	
	RadStd		0.04	0.12	0.31	0.40	0.55	0.50	0.37	0.36	0.27	0.11	0.06	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	+1.67	+1.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page